

Chapter 1

1.7. For each signal given below, determine all the values of the independent variable at which the even part of the signal is guaranteed to be zero.

- (a) $x_1[n] = u[n] - u[n-4]$ (b) $x_2(t) = \sin(\frac{1}{2}t)$
 (c) $x_3[n] = (\frac{1}{2})^n u[n-3]$ (d) $x_4(t) = e^{-5t} u(t+2)$

1.7. (a)

$$\varepsilon_e(x_1[n]) = \frac{1}{2} \{x_1[n] + x_1[-n]\} = \frac{1}{2} (u[n] - u[n-4] + u[-n] - u[-n-4])$$

Therefore, $\varepsilon_e(x_1[n])$ is zero for $|n| > 3$.

(b) Since $x_2(t)$ is an odd signal, $\varepsilon_e(x_2[t])$ is zero for all values of t .

$$(c) \quad \varepsilon_e(x_3[n]) = \frac{1}{2} \{x_3[n] + x_3[-n]\} = \frac{1}{2} \left[\left(\frac{1}{2}\right)^n u[n-3] - \left(\frac{1}{2}\right)^{-n} u[-n-3] \right]$$

Therefore, $\varepsilon_e(x_3[n])$ is zero when $|n| < 3$ and when $|n| \rightarrow \infty$.

$$(d) \quad \varepsilon_e(x_4(t)) = \frac{1}{2} (x_4(t) + x_4(-t)) = \frac{1}{2} [e^{-5t} u(t-2) + e^{5t} u(-t+2)]$$

Therefore, $\varepsilon_e(x_4(t))$ is zero only when $|t| \rightarrow \infty$.

注意看 c, d 题, 趋向于无穷的时候也是符合要求的

1.14. Consider a periodic signal

$$x(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ -2, & 1 < t < 2 \end{cases}$$

with period $T = 2$. The derivative of this signal is related to the “impulse train”

$$g(t) = \sum_{k=-\infty}^{\infty} \delta(t - 2k)$$

with period $T = 2$. It can be shown that

$$\frac{dx(t)}{dt} = A_1 g(t - t_1) + A_2 g(t - t_2).$$

Determine the values of A_1 , t_1 , A_2 , and t_2 .

1.14 The signal $x(t)$ and its derivative $g(t)$ are shown in Figure S1.14.

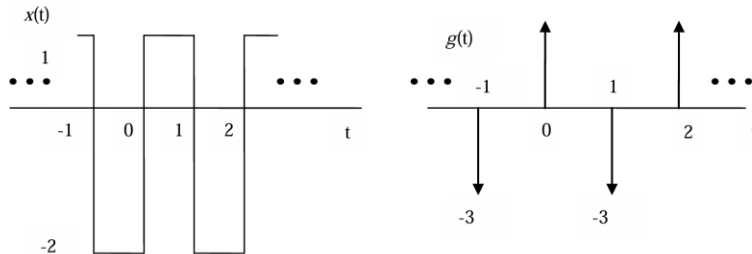


Figure S 1.14

Therefore

$$g(t) = 3 \sum_{k=-\infty}^{\infty} \delta(t - 2k) - 3 \sum_{k=-\infty}^{\infty} \delta(t - 2k - 1)$$

This implies that $A_1 = 3$, $t_1 = 0$, $A_2 = -3$, and $t_2 = 1$.

关注 A_1 A_2 的值, 是从前一次到后一次的改变值, 不是直接看 $x(t)$ 的值

1.15. Consider a system S with input $x[n]$ and output $y[n]$. This system is obtained through a series interconnection of a system S_1 followed by a system S_2 . The input-output relationships for S_1 and S_2 are

$$S_1: \quad y_1[n] = 2x_1[n] + 4x_1[n-1],$$

$$S_2: \quad y_2[n] = x_2[n-2] + \frac{1}{2}x_2[n-3],$$

where $x_1[n]$ and $x_2[n]$ denote input signals.

(a) Determine the input-output relationship for system S .

(b) Does the input-output relationship of system S change if the order in which S_1 and S_2 are connected in series is reversed (i.e., if S_2 follows S_1)?

(b) The input-output relationship does not change if the order in which S_1 and S_2 are connected series reversed. . We can easily prove this assuming that S_1 follows S_2 . In this case , the signal $x_1[n]$, which is the input to S_1 is the same as $y_2[n]$.

Therefore

$$\begin{aligned} y_1[n] &= 2x_1[n] + 4x_1[n-1] \\ &= 2y_2[n] + 4y_2[n-1] \\ &= 2\left(x_2[n-2] + \frac{1}{2}x_2[n-3]\right) + 4\left(x_2[n-3] + \frac{1}{2}x_2[n-4]\right) \\ &= 2x_2[n-2] + 5x_2[n-3] + 2x_2[n-4] \end{aligned}$$

The input-output relationship for S is once again

$$y[n] = 2x[n-2] + 5x[n-3] + 2x[n-4]$$

1.17. Consider a continuous-time system with input $x(t)$ and output $y(t)$ related by

$$y(t) = x(\sin(t)).$$

- (a) Is this system causal?
(b) Is this system linear?

1.17 (a) The system is not causal because the output $y(t)$ at some time may depend on future values of $x(t)$. For instance , $y(-\pi) = x(0)$.

(b) Consider two arbitrary inputs $x_1(t)$ and $x_2(t)$.

$$x_1(t) \rightarrow y_1(t) = x_1(\sin(t))$$

$$x_2(t) \rightarrow y_2(t) = x_2(\sin(t))$$

Let $x_3(t)$ be a linear combination of $x_1(t)$ and $x_2(t)$. That is , $x_3(t) = ax_1(t) + bx_2(t)$

Where a and b are arbitrary scalars .If $x_3(t)$ is the input to the given system ,then the corresponding output

$$\begin{aligned} y_3(t) &= x_3(\sin(t)) \\ &= ax_1(\sin(t)) + x_2(\sin(t)) \\ &= ay_1(t) + by_2(t) \end{aligned}$$

Therefore , the system is linear.

1.19. For each of the following input-output relationships, determine whether the corresponding system is linear, time invariant or both.

- (a) $y(t) = t^2 x(t-1)$ (b) $y[n] = x^2[n-2]$
(c) $y[n] = x[n+1] - x[n-1]$ (d) $y[n] = \mathcal{O}\{x(t)\}$

1.19 (a) (i) Consider two arbitrary inputs $x_1(t)$ and $x_2(t)$.

$$x_1(t) \rightarrow y_1(t) = t^2 x_1(t-1)$$

$$x_2(t) \rightarrow y_2(t) = t^2 x_2(t-1)$$

Let $x_3(t)$ be a linear combination of $x_1(t)$ and $x_2(t)$. That is

$$x_3(t) = ax_1(t) + bx_2(t)$$

where a and b are arbitrary scalars. If $x_3(t)$ is the input to the given system, then the corresponding output

$$\begin{aligned} y_3(t) &= t^2 x_3(t-1) \\ &= t^2 (ax_1(t-1) + bx_2(t-1)) \\ &= ay_1(t) + by_2(t) \end{aligned}$$

Therefore , the system is linear.

(ii) Consider an arbitrary inputs $x_1(t)$. Let $y_1(t) = t^2 x_1(t-1)$

be the corresponding output .Consider a second input $x_2(t)$ obtained by shifting $x_1(t)$ in time:

$$x_2(t) = x_1(t-t_0)$$

The output corresponding to this input is $y_2(t) = t^2 x_2(t-1) = t^2 x_1(t-1-t_0)$

Also note that

$$y_1(t-t_0) = (t-t_0)^2 x_1(t-1-t_0) \neq y_2(t)$$

Therefore the system is not time-invariant.

(d) (i) Consider two arbitrary inputs $x_1(t)$ and $x_2(t)$.

$$x_1(t) \rightarrow y_1(t) = \text{Od}\{x_1(t)\}$$

$$x_2(t) \rightarrow y_2(t) = \text{Od}\{x_2(t)\}$$

Let $x_3(t)$ be a linear combination of $x_1(t)$ and $x_2(t)$. That is $x_3(t) = a x_1(t) + b x_2(t)$

where a and b are arbitrary scalars. If $x_3(t)$ is the input to the given system, then the corresponding output

$$\begin{aligned} y_3(t) \text{ is } y_3(t) &= \text{Od}\{x_3(t)\} \\ &= \text{Od}\{a x_1(t) + b x_2(t)\} \\ &= a \text{Od}\{x_1(t)\} + b \text{Od}\{x_2(t)\} = a y_1(t) + b y_2(t) \end{aligned}$$

Therefore the system is linear.

(ii) Consider an arbitrary inputs $x_1(t)$. Let

$$y_1(t) = \text{Od}\{x_1(t)\} = \frac{x_1(t) - x_1(-t)}{2}$$

be the corresponding output. Consider a second input $x_2(t)$ obtained by shifting $x_1(t)$ in time:

$$x_2(t) = x_1(t - t_0)$$

The output corresponding to this input is

$$\begin{aligned} y_2(t) &= \text{Od}\{x_2(t)\} = \frac{x_2(t) - x_2(-t)}{2} \\ &= \frac{x_1(t - t_0) - x_1(-t - t_0)}{2} \end{aligned}$$

Also note that

$$y_1(t - t_0) = \frac{x_1(t - t_0) - x_1(-t - t_0)}{2} \neq y_2(t)$$

Therefore the system is not time-invariant.

1.27. In this chapter, we introduced a number of general properties of systems. In particular, a system may or may not be

- (1) Memoryless
- (2) Time invariant
- (3) Linear
- (4) Causal
- (5) Stable

Determine which of these properties hold and which do not hold for each of the following continuous-time systems. Justify your answers. In each example, $y(t)$ denotes the system output and $x(t)$ is the system input.

第 5 个最阴影:

$$\int_0^t x(\tau) d\tau$$

看起来像普通积分, 但下限固定在绝对时间 0, 所以时变。

而:

$$\int_{t-2}^t x(\tau) d\tau$$

窗口随 t 整体移动, 所以时不变。

1.30. Determine if each of the following systems is invertible. If it is, construct the inverse system. If it is not, find two input signals to the system that have the same output.

(a) $y(t) = x(t - 4)$

(b) $y(t) = \cos[x(t)]$

(c) $y[n] = nx[n]$

(d) $y(t) = \int_{-\infty}^t x(\tau) d\tau$

(e) $y[n] = \begin{cases} x[n-1], & n \geq 1 \\ 0, & n = 0 \\ x[n], & n \leq -1 \end{cases}$

(f) $y[n] = x[n]x[n-1]$

(g) $y[n] = x[1-n]$

(h) $y(t) = \int_{-\infty}^t e^{-(t-\tau)} x(\tau) d\tau$

(i) $y[n] = \sum_{k=-\infty}^n (\frac{1}{2})^{n-k} x[k]$

(j) $y(t) = \frac{dx(t)}{dt}$

(k) $y[n] = \begin{cases} x[n+1], & n \geq 0 \\ x[n], & n \leq -1 \end{cases}$

(l) $y(t) = x(2t)$

(m) $y[n] = x[2n]$

(n) $y[n] = \begin{cases} x[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$

要注意离散型的有无索引缺失导致的

1.31. In this problem, we illustrate one of the most important consequences of the properties of linearity and time invariance. Specifically, once we know the response of a linear system or a linear time-invariant (LTI) system to a single input or the responses to several inputs, we can directly compute the responses to many other

input signals. Much of the remainder of this book deals with a thorough exploitation of this fact in order to develop results and techniques for analyzing and synthesizing LTI systems.

- (a) Consider an LTI system whose response to the signal $x_1(t)$ in Figure P1.31(a) is the signal $y_1(t)$ illustrated in Figure P1.31(b). Determine and sketch carefully the response of the system to the input $x_2(t)$ depicted in Figure P1.31(c).
 (b) Determine and sketch the response of the system considered in part (a) to the input $x_3(t)$ shown in Figure P1.31(d).

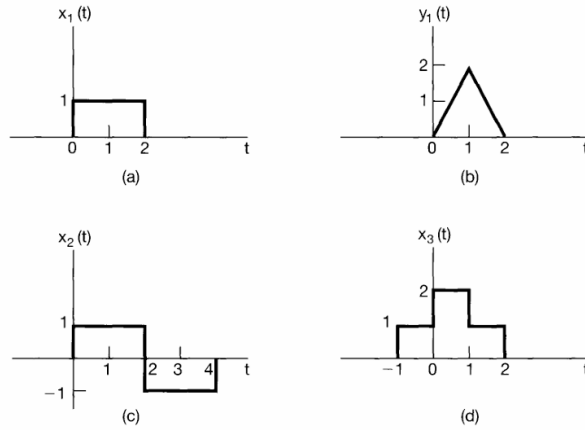


Figure P1.31

B 题的拆法

注意这种题目，只能有 amplitude scaling 和 time shifting 两种，不能使用 time scaling 和 time reversal

- 1.36. (a) If $x[n]$ is periodic $e^{j\omega_0(n+N)T}$, where $\omega_0 = 2\pi/T_0$. This implies that

$$\frac{2\pi}{T_0} NT = 2\pi k \Rightarrow \frac{T}{T_0} = \frac{k}{N} = \text{a rational number}.$$

- (b) $T/T_0 = p/q$ then $x[n] = e^{j2\pi n(p/q)}$. The fundamental period is $q/\gcd(p, q)$ and the fundamental frequency is

$$\frac{2\pi}{q} \gcd(p, q) = \frac{2\pi}{p} \frac{p}{q} \gcd(p, q) = \frac{\omega_0}{p} \gcd(p, q) = \frac{\omega_0 T}{p} \gcd(p, q).$$

- (c) $p/\gcd(p, q)$ periods of $x(t)$ are needed.

- 1.36. Let $x(t)$ be the continuous-time complex exponential signal

$$x(t) = e^{j\omega_0 t}$$

with fundamental frequency ω_0 and fundamental period $T_0 = 2\pi/\omega_0$. Consider the discrete-time signal obtained by taking equally spaced samples of $x(t)$ —that is,

$$x[n] = x(nT) = e^{j\omega_0 nT}.$$

- (a) Show that $x[n]$ is periodic if and only if T/T_0 is a rational number—that is, if and only if some multiple of the sampling interval *exactly equals* a multiple of the period of $x(t)$.
 (b) Suppose that $x[n]$ is periodic—that is, that

$$\frac{T}{T_0} = \frac{p}{q}, \quad (\text{P1.36-1})$$

where p and q are integers. What are the fundamental period and fundamental frequency of $x[n]$? Express the fundamental frequency as a fraction of $\omega_0 T$.

- (c) Again assuming that T/T_0 satisfies eq. (P1.36-1), determine precisely how many periods of $x(t)$ are needed to obtain the samples that form a single period of $x[n]$.

- 1.37.** An important concept in many communications applications is the *correlation* between two signals. In the problems at the end of Chapter 2, we will have more to say about this topic and will provide some indication of how it is used in practice. For now, we content ourselves with a brief introduction to correlation functions and some of their properties.

Let $x(t)$ and $y(t)$ be two signals; then the *correlation function* is defined as

$$\phi_{xy}(t) = \int_{-\infty}^{\infty} x(t + \tau)y(\tau)d\tau.$$

The function $\phi_{xx}(t)$ is usually referred to as the *autocorrelation function* of the signal $x(t)$, while $\phi_{xy}(t)$ is often called a *cross-correlation function*.

- (a) What is the relationship between $\phi_{xy}(t)$ and $\phi_{yx}(t)$?
 (b) Compute the odd part of $\phi_{xx}(t)$.
 (c) Suppose that $y(t) = x(t + T)$. Express $\phi_{xy}(t)$ and $\phi_{yy}(t)$ in terms of $\phi_{xx}(t)$.

- 1.37.** (a) From the definition of $\phi_{xy}(t)$. We have

$$\begin{aligned}\phi_{xy}(t) &= \int_{-\infty}^{\infty} x(t + \tau)y(\tau)d\tau \\ &= \int_{-\infty}^{\infty} y(-t + \tau)x(\tau)d\tau\end{aligned}$$

- (b) Note from $\phi_{yx}(-t)$. part(a) that $\phi_{xx}(t) = \phi_{xx}(-t)$. This implies that $\phi_{xy}(t)$ is even. Therefore,

the odd part of $\phi_{xx}(t)$ is zero.

- (c) Here, $\phi_{xy}(t) = \phi_{xx}(t - T)$ and $\phi_{yy}(t) = \phi_{xx}(t)$.

一步步写出来，然后做替换

- 1.43.** (a) Consider a time-invariant system with input $x(t)$ and output $y(t)$. Show that if $x(t)$ is periodic with period T , then so is $y(t)$. Show that the analogous result also holds in discrete time.
 (b) Give an example of a time-invariant system and a nonperiodic input signal $x(t)$ such that the corresponding output $y(t)$ is periodic.

- 1.43.** (a) We have

$$x(t) \xrightarrow{S} y(t)$$

Since S is time-invariant.

$$x(t - T) \xrightarrow{S} y(t - T)$$

Now if $x(t)$ is periodic with period T , $x(t) = x(t - T)$. Therefore, we may conclude that $y(t) = y(t - T)$. This implies that $y(t)$ is also periodic with T . A similar argument may be made in discrete time.

(b)

- 1.44.** (a) Show that causality for a continuous-time linear system is equivalent to the following statement:

For any time t_0 and any input $x(t)$ such that $x(t) = 0$ for $t < t_0$, the corresponding output $y(t)$ must also be zero for $t < t_0$.

The analogous statement can be made for a discrete-time linear system.

- (b) Find a nonlinear system that satisfies the foregoing condition but is not causal.
 (c) Find a nonlinear system that is causal but does not satisfy the condition.
 (d) Show that invertibility for a discrete-time linear system is equivalent to the following statement:

The only input that produces $y[n] = 0$ for all n is $x[n] = 0$ for all n .

The analogous statement is also true for a continuous-time linear system.

- (e) Find a nonlinear system that satisfies the condition of part (d) but is not invertible.

(a) 对线性系统，causal 等价于：输入在 $t < t_0$ 为零，则输出也在 $t < t_0$ 为零

(b) $y(t) = x(t)x(t + 1)$ 满足 foregoing condition，但 non-causal。

(c) $y(t) = x(t) + 1$ causal，但不满足 foregoing condition。

(d) 对线性系统，invertible 等价于：唯一产生零输出的是零输入

$$(e) \ y[n] = x^2[n]$$

Chapter 2

2.3. Consider an input $x[n]$ and a unit impulse response $h[n]$ given by

$$x[n] = \left(\frac{1}{2}\right)^{n-2} u[n-2],$$

$$h[n] = u[n+2].$$

Determine and plot the output $y[n] = x[n] * h[n]$.

注意是否离散

Problem 4

Find the impulse response of the following two systems *using the time-domain five-step method* demonstrated in class. **Do not use the Laplace transform.**

- a) $\ddot{y}(t) + 2\dot{y}(t) + y(t) = \dot{x}(t) + 2x(t)$
- b) $\ddot{y}(t) + 4\dot{y}(t) + 5y(t) + 2y(t) = \dot{x}(t) + 3x(t)$

2.18. Consider a causal LTI system whose input $x[n]$ and output $y[n]$ are related by the difference equation

$$y[n] = \frac{1}{4}y[n-1] + x[n].$$

Determine $y[n]$ if $x[n] = \delta[n-1]$.

这种离散的题目，直接多推导几项就能找出规律

2.20. Evaluate the following integrals:

- (a) $\int_{-\infty}^{\infty} u_0(t) \cos(t) dt$
- (b) $\int_0^5 \sin(2\pi t) \delta(t+3) dt$
- (c) $\int_{-5}^5 u_1(1-\tau) \cos(2\pi\tau) d\tau$

U 下面的角标是因为阶跃函数有歧义，下面的角标就是 $t=0$ 时刻的取值

Chapter 3

3.10. Let $x[n]$ be a real and odd periodic signal with period $N = 7$ and Fourier coefficients a_k . Given that

$$a_{15} = j, a_{16} = 2j, a_{17} = 3j,$$

determine the values of a_0 , a_{-1} , a_{-2} , and a_{-3} .

离散信号的傅里叶级数 a_k 呈周期性，周期和信号的周期保持一致

3.13. Consider a continuous-time LTI system whose frequency response is

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt = \frac{\sin(4\omega)}{\omega}.$$

If the input to this system is a periodic signal

$$x(t) = \begin{cases} 1, & 0 \leq t < 4 \\ -1, & 4 \leq t < 8 \end{cases}$$

with period $T = 8$, determine the corresponding system output $y(t)$.

3.13 Let us first evaluate the Fourier series coefficients of $x(t)$. Clearly, since $x(t)$ is real and odd, a_k is purely imaginary and odd. Therefore, $a_0 = 0$. Now,

$$\begin{aligned} a_k &= \frac{1}{8} \int_0^8 x(t) e^{-j(2\pi/8)kt} dt \\ &= \frac{1}{8} \int_0^4 e^{-j(2\pi/8)kt} dt - \frac{1}{8} \int_0^4 e^{-j(2\pi/8)kt} dt \\ &= \frac{1}{j\pi k} [1 - e^{-j\pi k}] \end{aligned}$$

Clearly, the above expression evaluates to zero for all even values of k . Therefore,

$$a_k = \begin{cases} 0, & k = 0, \pm 2, \pm 4, \dots \\ \frac{2}{j\pi k}, & k = \pm 1, \pm 3, \pm 5, \dots \end{cases}$$

When $x(t)$ is passed through an LTI system with frequency response $H(j\omega)$, the output $y(t)$ is given by (see Section 3.8)

$$y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$$

Where $\omega_0 = \frac{2\pi}{T} = \frac{\pi}{4}$. Since a_k is non zero only for odd values of k , we need to evaluate the above summation only for odd k . Furthermore, note that

$$H(jk\omega_0) = H(jk(\pi/4)) = \frac{\sin(k\pi)}{k(\pi/4)}$$

is always zero for odd values of k . Therefore,

$$y(t) = 0.$$

Frequency response 中的 $j\omega$ 中的 ω 用 $\omega_0 k$ 带入其中

3.17. Consider three continuous-time systems S_1 , S_2 , and S_3 whose responses to a complex exponential input e^{j5t} are specified as

$$\begin{aligned} S_1 : e^{j5t} &\longrightarrow te^{j5t}, \\ S_2 : e^{j5t} &\longrightarrow e^{j5(t-1)}, \\ S_3 : e^{j5t} &\longrightarrow \cos(5t). \end{aligned}$$

For each system, determine whether the given information is sufficient to conclude that the system is definitely *not* LTI.

3.17 (a) Since complex exponentials are Eigen functions of LTI systems, the input $x_1(t) = e^{j5t}$ has to produce an output of the form Ae^{j5t} , where A is a complex constant. By clearly, in this case the output is not of this form. Therefore, system S_1 is definitely not LTI.

(b) This system may be LTI because it satisfies the Eigen function property of LTI systems.

(c) In the case, the output is of the form $y_3(t) = (1/2)e^{j5t} + (1/2)e^{-j5t}$. Clearly, the output contains a complex exponential with frequency -5 which was not present in the input $x_3(t)$. We know that an LTI system can never produce a complex exponential of frequency -5 unless there was a complex exponential of same frequency at its input. Since this is not the case in this problem, S_3 is definitely not LTI.

核心思想：LTI 系统不能产生输入信号没有的新频率

3.20. Consider a causal LTI system implemented as the RLC circuit shown in Figure P3.20. In this circuit, $x(t)$ is the input voltage. The voltage $y(t)$ across the capacitor is considered the system output.

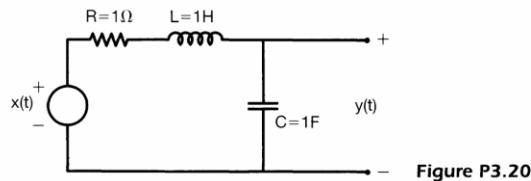


Figure P3.20

- Find the differential equation relating $x(t)$ and $y(t)$.
- Determine the frequency response of this system by considering the output of the system to inputs of the form $x(t) = e^{j\omega t}$.
- Determine the output $y(t)$ if $x(t) = \sin(t)$.

(b) 求 frequency response $H(j\omega)$

令输入为：

$$x(t) = e^{j\omega t}$$

对于 LTI 系统，输出形式为：

$$y(t) = H(j\omega)e^{j\omega t}$$

代入微分方程：

$$y''(t) + y'(t) + y(t) = x(t)$$

先求导：

$$y'(t) = j\omega H(j\omega)e^{j\omega t}$$

$$y''(t) = (j\omega)^2 H(j\omega)e^{j\omega t}$$

代入：

$$(j\omega)^2 H(j\omega)e^{j\omega t} + j\omega H(j\omega)e^{j\omega t} + H(j\omega)e^{j\omega t} = e^{j\omega t}$$

提取：

$$H(j\omega)e^{j\omega t}[(j\omega)^2 + j\omega + 1] = e^{j\omega t}$$

消去 $e^{j\omega t}$ ：

$$H(j\omega)[(j\omega)^2 + j\omega + 1] = 1$$

所以：

$$H(j\omega) = \frac{1}{(j\omega)^2 + j\omega + 1}$$

(c) The signal $x(t)$ is periodic with period 2π . Since $x(t)$ can be expressed in the form

$$x(t) = \frac{1}{2j} e^{j(2\pi/2\pi)t} - \frac{1}{2j} e^{-j(2\pi/2\pi)t}$$

the non-zero Fourier series coefficients of $x(t)$ are

$$a_1 = a_{-1} = \frac{1}{2j}$$

Using the results derived in Section 3.8 (see eq.(3.124)), we have

$$\begin{aligned} y(t) &= a_1 H(j) e^{jt} - a_{-1} H(-j) e^{-jt} \\ &= (1/2j) \left(\frac{1}{j} e^{jt} - \frac{1}{-j} e^{-jt} \right) \\ &= (-1/2) (e^{jt} + e^{-jt}) \\ &= -\cos(t) \end{aligned}$$

3.23. In each of the following, we specify the Fourier series coefficients of a continuous-time signal that is periodic with period 4. Determine the signal $x(t)$ in each case.

$$(a) a_k = \begin{cases} 0, & k = 0 \\ (j)^k \frac{\sin k\pi/4}{k\pi}, & \text{otherwise} \end{cases}$$

$$(b) a_k = (-1)^k \frac{\sin k\pi/8}{2k\pi}, \quad a_0 = \frac{1}{16}$$

$$(c) a_k = \begin{cases} jk, & |k| < 3 \\ 0, & \text{otherwise} \end{cases}$$

$$(d) a_k = \begin{cases} 1, & k \text{ even} \\ 2, & k \text{ odd} \end{cases}$$

一个常用模板

如果一个周期矩形脉冲：

$$x(t) = A, c - \frac{W}{2} < t < c + \frac{W}{2}$$

周期为 $T = 4$ ，其他地方为 0，那么它的傅里叶系数是：

$$a_k = A \frac{\sin(k\omega_0 W/2)}{k\pi} e^{-jk\omega_0 c}, k \neq 0$$

其中：

$$\omega_0 = \frac{\pi}{2}$$

而：

$$a_0 = \frac{AW}{T}$$

Consider a continuous-time periodic signal $x(t)$ with period $T = 2$ and Fourier coefficients a_k . Determine $x(t)$ with the following information:

$$\diamond a_k = a_{k+2} \text{ for all } k$$

$$\diamond \int_{-0.5}^{0.5} x(t) dt = 2$$

$$\diamond \int_{0.5}^{1.5} x(t) dt = 1$$

把偶数项和奇数项分开：

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\pi t}$$

偶数项令 $k = 2m$ ：

$$\sum_{m=-\infty}^{\infty} A e^{j2m\pi t} = A \sum_{m=-\infty}^{\infty} e^{j2\pi m t}$$

奇数项令 $k = 2m + 1$ ：

$$\sum_{m=-\infty}^{\infty} B e^{j(2m+1)\pi t} = B e^{j\pi t} \sum_{m=-\infty}^{\infty} e^{j2\pi m t}$$

所以：

$$x(t) = (A + B e^{j\pi t}) \sum_{m=-\infty}^{\infty} e^{j2\pi m t}$$

关键现在是理解：

$$\sum_{m=-\infty}^{\infty} e^{j2\pi m t}$$

它不是普通函数。它等价于周期为 1 的冲激串：

$$\sum_{m=-\infty}^{\infty} e^{j2\pi m t} = \sum_{n=-\infty}^{\infty} \delta(t - n)$$

为什么？因为周期为 1 的冲激串

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - n)$$

在一个周期内只有一个单位冲激。它的傅里叶系数是：

$$c_m = \int_{-0.5}^{0.5} \delta(t) e^{-j2\pi m t} dt = 1$$

所以它的傅里叶级数就是：

$$p(t) = \sum_{m=-\infty}^{\infty} 1 \cdot e^{j2\pi m t}$$

也就是：

$$\sum_{m=-\infty}^{\infty} e^{j2\pi mt} = \sum_{n=-\infty}^{\infty} \delta(t-n)$$

3.22. Determine the Fourier series representations for the following signals:

(a) Each $x(t)$ illustrated in Figure P3.22(a)–(f).

(b) $x(t)$ periodic with period 2 and

$$x(t) = e^{-t} \quad \text{for} \quad -1 < t < 1$$

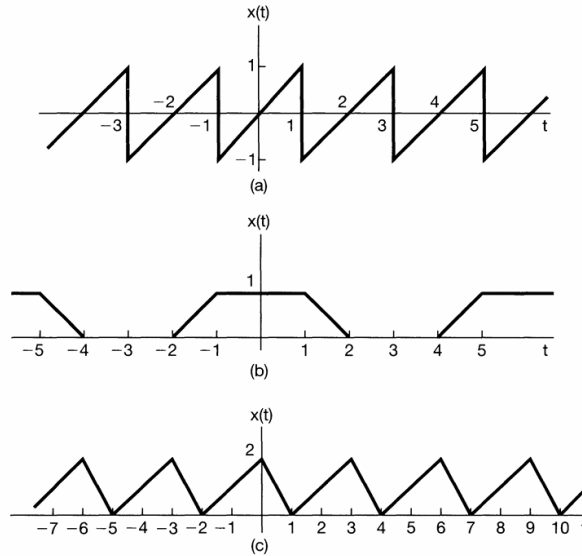


Figure P3.22

图 (c) 是周期三角波，根据图像可取一个周期为 $[-2, 1]$ ，周期为

$$T = 3, \omega_0 = \frac{2\pi}{T} = \frac{2\pi}{3}$$

一个周期内：

$$x(t) = \begin{cases} t + 2, & -2 < t < 0 \\ 2 - 2t, & 0 < t < 1 \end{cases}$$

连续时间复指数傅里叶级数形式：

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

其中

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$$

所以

$$a_k = \frac{1}{3} \left[\int_{-2}^0 (t+2) e^{-jk\frac{2\pi}{3}t} dt + \int_0^1 (2-2t) e^{-jk\frac{2\pi}{3}t} dt \right]$$

结果

当 $k = 0$

$$a_0 = \frac{1}{3} \cdot \text{area of one triangle}$$

三角形底为 3，高为 2，面积：

$$\frac{1}{2} \cdot 3 \cdot 2 = 3$$

所以

$$a_0 = 1$$

当 $k \neq 0$

$$a_k = \frac{9}{4\pi^2 k^2} \left(1 - e^{-j\frac{2\pi}{3}k} \right)$$

因此完整傅里叶级数为：

$$x(t) = 1 + \sum_{\substack{k=-\infty \\ k \neq 0}}^{\infty} \frac{9}{4\pi^2 k^2} \left(1 - e^{-j\frac{2\pi}{3}k} \right) e^{jk\frac{2\pi}{3}t}$$

注意这一步：**5. 利用 k 是整数进行化简**

因为

$$\alpha = k \frac{2\pi}{3}$$

所以

$$3\alpha = 2\pi k$$

于是

$$e^{j3\alpha} = e^{j2\pi k} = 1$$

因此

$$e^{j2\alpha} = e^{-j\alpha}$$

原因是：

$$e^{j2\alpha} = e^{-j\alpha} e^{j3\alpha} = e^{-j\alpha}$$

所以

$$e^{j2\alpha} + 2e^{-j\alpha} - 3 = 3e^{-j\alpha} - 3 = 3(e^{-j\alpha} - 1)$$

于是

$$I_1 + I_2 = \frac{3(e^{-j\alpha} - 1)}{(-j\alpha)^2}$$

又因为

$$(-j\alpha)^2 = -\alpha^2$$

所以

$$I_1 + I_2 = \frac{3(e^{-j\alpha} - 1)}{-\alpha^2} = \frac{3(1 - e^{-j\alpha})}{\alpha^2}$$

凡是傅里叶级数积分后出现相差一个周期相位的指数项，都可以这样化简

Determine whether the following systems are 1) static or dynamic; 2) causal or non-causal; 3) invertible or non-invertible; 4) stable or unstable. Justify your answers.

$$(a) y(t) = x(t-2) + x(2-t)$$

$$(b) y(t) = [\cos(3t)]x(t)$$

$$(c) y(t) = \int_{-\infty}^{2t} x(\tau) d\tau$$

$$(d) y(t) = \begin{cases} 0, & t < 0 \\ x(t) + x(t-2), & t \geq 0 \end{cases}$$

$$(e) y(t) = \begin{cases} 0, & x(t) < 0 \\ x(t) + x(t-2), & x(t) \geq 0 \end{cases}$$

$$(f) y(t) = x(t/3)$$

$$(g) y(t) = \frac{dx(t)}{dt}$$

D: non-invertible; time-variant; linear

有信息丢失； 时间边界 $t=0$ 不能改变，所以时变

E: non-invertible; time-invariant; non-linear

有信息丢失； 时间边界会随着一起改变；使用系数为负作为反例

Find the Fourier series expansion of the following signal:

$$x(t) = \cos \frac{2\pi}{T}t + \left| \cos \frac{2\pi}{T}t \right|$$

1. 第一部分: $\cos(\omega_0 t)$

$$\cos(\omega_0 t) = \frac{1}{2}e^{j\omega_0 t} + \frac{1}{2}e^{-j\omega_0 t}$$

所以它贡献:

$$a_1 = \frac{1}{2}, a_{-1} = \frac{1}{2}$$

2. 第二部分: $|\cos(\omega_0 t)|$

标准展开为:

$$|\cos \theta| = \frac{2}{\pi} + \frac{4}{\pi} \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{4m^2 - 1} \cos(2m\theta)$$

令:

$$\theta = \omega_0 t$$

得到:

$$|\cos(\omega_0 t)| = \frac{2}{\pi} + \frac{4}{\pi} \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{4m^2 - 1} \cos(2m\omega_0 t)$$

再把余弦转成复指数:

$$\cos(2m\omega_0 t) = \frac{1}{2}e^{j2m\omega_0 t} + \frac{1}{2}e^{-j2m\omega_0 t}$$

所以 $|\cos(\omega_0 t)|$ 的复指数系数是:

$$a_0 = \frac{2}{\pi}$$

$$a_{\pm 2m} = \frac{2(-1)^{m+1}}{\pi(4m^2 - 1)}, m = 1, 2, 3, \dots$$

奇数项为 0。